

Lecture 6 — Scarcity and the Value of Capacity: Littlewood's Rule

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6.1 From price discrimination to capacity allocation

So far we have mostly asked: what price should a seller charge? But why should every customer necessarily face the same price? Economists distinguish three broad forms of *price discrimination*:

First degree.

The seller tailors the price to an individual customer's willingness to pay. In the idealized full-information setting of Lecture 1, if buyer i has known value v_i , the seller could charge $p_i = v_i$. This is perfect, or first-degree, price discrimination.

Second degree.

The seller offers a menu of products, contracts, or prices, and customers self-select. Examples include quantity discounts, basic versus premium versions, refundable versus nonrefundable tickets, advance-purchase restrictions, and different levels of flexibility.

Third degree.

The seller observes a customer segment and charges different prices to different groups, such as students, seniors, geographic markets, or customer categories.

Later lectures will return to observable segmentation, versioning and self-selection, and mechanism-design perspectives on hidden information. Here this taxonomy is motivation for a different question.

Airline fare differentiation is often closest to second-degree price discrimination. An airline might offer a restricted, advance-purchase fare for \$100 and a flexible, late, or refundable fare for \$250, allowing customers to self-select. Not every fare-class system is purely second-degree discrimination, however, and the abstract model below does not model the self-selection mechanism. It simply takes two revenue classes as given. If the classes instead represented observed customer types receiving different prices, the abstraction would resemble third-degree discrimination.

The distinction for this lecture is

price discrimination	→	which price or fare class?
revenue management	→	how much scarce capacity should each class receive?

Airline seats, hotel-room nights, advertising impressions, and event tickets have limited capacity, requests that arrive over time, and inventory that expires. Our central question is therefore:

Why might it be optimal to reject a profitable customer today?

6.2 One seat and the value of waiting

Suppose one seat remains. A low-fare customer is present now and pays $r_L = \$100$. Tomorrow a high-fare customer pays $r_H = \$250$ if they arrive, and they arrive with probability $q = 1/2$. The two choices have values

$$\begin{array}{l|l} \text{accept now} & 100 \\ \text{reject and protect the seat} & \frac{1}{2}(250) = 125. \end{array}$$

It is optimal to reject the low-fare customer even though the sale would produce positive revenue.

A positive-revenue customer can still be worth rejecting because capacity has an opportunity cost.

More generally, with one seat, accept the low fare if and only if

$$r_L \geq r_H \mathbb{P}(\text{high-fare demand arrives}). \quad (6.1)$$

The right-hand side is the expected displacement value of preserving the seat. This is Littlewood's rule in its simplest form.

This also sharpens the full-information discussion from Lecture 1. There, with known customers and no future uncertainty, a positive-revenue sale looked attractive. Here accepting one customer changes the capacity available to future customers. The comparison is not revenue now versus zero, but revenue now versus the opportunity cost of capacity:

Capacity is valuable because accepting one customer removes an option from the future.

6.3 Littlewood's two-stage capacity-allocation problem

A seller has C identical units of capacity. There are two customer classes:

- low-fare customers pay r_L and arrive early;
- high-fare customers pay $r_H > r_L$ and arrive later.

The later high-fare demand is the integer-valued random variable D_H . The seller knows its distribution but not its realization.

Suppose an early low-fare customer arrives when y units remain. Accepting produces r_L now, but leaves only $y - 1$ units for high-fare demand. Rejecting preserves all y units.

The expected high-fare revenue with y protected units is

$$r_H \mathbb{E}[\min\{D_H, y\}].$$

The value of protecting the y th unit is therefore

$$\begin{aligned} r_H \mathbb{E}[\min\{D_H, y\} - \min\{D_H, y - 1\}] \\ = r_H \mathbb{P}(D_H \geq y). \end{aligned}$$

The equality follows because the y th unit is used precisely when high-fare demand reaches at least y . Thus the many-seat comparison is the same as in the one-seat example:

$$\underbrace{r_L}_{\text{revenue for certain now}} \quad \text{versus} \quad \underbrace{r_H \mathbb{P}(D_H \geq y)}_{\text{expected displacement value of the marginal protected unit}}.$$

Proposition 1 (Littlewood's rule). *When y units remain, accept an early low-fare request if and only if*

$$r_L \geq r_H \mathbb{P}(D_H \geq y). \quad (6.2)$$

Equivalently, protect units for high-fare demand as long as their expected marginal value exceeds the low fare.

This is a capacity-allocation statement given the two revenue classes, not a model of how those classes were created. It is a dynamic capacity-control principle, even though the fares are fixed:

$\text{price of capacity} = \text{expected revenue displaced by using it now.}$

6.4 A numerical Littlewood example

Let

$$r_L = 100, \quad r_H = 250, \quad D_H \sim \text{Binomial}(3, 1/2).$$

The relevant tail probabilities are

y	1	2	3
$\mathbb{P}(D_H \geq y)$	7/8	1/2	1/8
$r_H \mathbb{P}(D_H \geq y)$	218.75	125	31.25.

Thus:

- with one unit remaining, reject the low fare: $218.75 > 100$;
- with two units remaining, reject the low fare: $125 > 100$;
- with three units remaining, accept the low fare: $31.25 < 100$.

The optimal protection level is two units. Equivalently,

$$y^* = \max \left\{ y : \mathbb{P}(D_H \geq y) > \frac{r_L}{r_H} \right\} = 2.$$

More generally, with the convention that ties may be resolved by accepting the low fare, the protection level is

$$y^* = \max \left\{ y \leq C : \mathbb{P}(D_H \geq y) > \frac{r_L}{r_H} \right\}.$$

If $F_H(d) = \mathbb{P}(D_H \leq d)$ and $y^* < C$, this is the discrete critical-fractile condition

$$F_H(y^* - 1) < 1 - \frac{r_L}{r_H} \quad \text{and} \quad F_H(y^*) \geq 1 - \frac{r_L}{r_H}.$$

Thus the protection level is a quantile of high-fare demand, with the exact inequalities determined by discreteness and the chosen tie-breaking rule.

Remark 6.1 (Protection level versus total capacity). *The high-fare demand distribution and fare ratio determine the number protected. Total capacity still determines the corresponding low-fare booking limit, $C - y^*$.*

6.5 The dynamic-programming viewpoint

Littlewood's model separates the two customer classes in time. We now allow different types to arrive in any period.

There are t periods remaining and b units of capacity. In a period:

- a type- j request arrives with probability λ_j and offers reward r_j ;
- no request arrives with probability λ_0 ;
- at most one unit can be sold;
- accept/reject decisions are irrevocable.

Our time index counts down: $v_t(b)$ is the maximum expected revenue with t *periods remaining*, including the next period in which a request may arrive, and with b units remaining. After the current decision, the continuation problem therefore has $t - 1$ periods. The boundary conditions are

$$v_0(b) = 0, \quad v_t(0) = 0.$$

The Bellman equation is

$$v_t(b) = \lambda_0 v_{t-1}(b) + \sum_j \lambda_j \max \{v_{t-1}(b), r_j + v_{t-1}(b-1)\}. \quad (6.3)$$

The two terms inside the maximum are:

$$\begin{aligned} \text{reject: } & v_{t-1}(b), \\ \text{accept: } & r_j + v_{t-1}(b-1). \end{aligned}$$

Therefore type j is accepted precisely when

$$r_j \geq v_{t-1}(b) - v_{t-1}(b-1). \quad (6.4)$$

Define the marginal value of capacity

$$\delta_{t-1}(b) = v_{t-1}(b) - v_{t-1}(b-1).$$

Then the optimal policy is simply

accept type j if $r_j \geq \delta_{t-1}(b)$.

6.6 Executing a small dynamic program

Consider a horizon of three periods and capacity at most two. In each period:

arrival	H	L	none
probability	0.3	0.5	0.2
reward	10	4	0.

We compute the value function backward.

6.6.1 One period remaining

With positive capacity, accept either customer:

$$v_1(1) = v_1(2) = 0.3(10) + 0.5(4) = 5.$$

6.6.2 Two periods remaining

With one unit, its marginal continuation value is $v_1(1) - v_1(0) = 5$. Accept H but reject L :

$$v_2(1) = 0.3(10) + 0.5(5) + 0.2(5) = 6.5.$$

With two units, using one unit costs no future revenue because $v_1(2) - v_1(1) = 0$. Accept both types:

$$v_2(2) = 0.3(10 + 5) + 0.5(4 + 5) + 0.2(5) = 10.$$

6.6.3 Three periods remaining

With one unit, the marginal continuation value is 6.5, so reject L :

$$v_3(1) = 0.3(10) + 0.5(6.5) + 0.2(6.5) = 7.55.$$

With two units, the marginal continuation value is

$$v_2(2) - v_2(1) = 10 - 6.5 = 3.5.$$

Now the low reward 4 exceeds the opportunity cost, so accept L :

$$v_3(2) = 0.3(10 + 6.5) + 0.5(4 + 6.5) + 0.2(10) = 12.2.$$

Collecting the values:

	$b = 0$	$b = 1$	$b = 2$
$t = 0$	0	0	0
$t = 1$	0	5	5
$t = 2$	0	6.5	10
$t = 3$	0	7.55	12.2

The low-fare decision depends on both time and capacity. We reject a low request when capacity is scarce relative to the remaining opportunities, but accept it when capacity is plentiful.

6.7 What the exact DP teaches us

This is the small-system regime: the state can be enumerated, so dynamic programming gives the exact marginal value of capacity. The DP provides three ideas that survive when the system grows and the recursion itself becomes impossible to compute.

6.7.1 1. Capacity has a state-dependent price

The quantity

$$\delta_t(b) = v_t(b) - v_t(b-1)$$

is the opportunity cost of one unit. It is high when capacity is scarce and future opportunities are valuable.

6.7.2 2. The relevant comparison is reward versus displacement

A sale is profitable in isolation whenever $r_j > 0$. It is desirable dynamically only when

$$r_j \geq \text{future value displaced.}$$

This is why “accept every profitable customer” is generally wrong.

6.7.3 3. The DP is an exact benchmark, not always a practical algorithm

For one resource with capacity C , the DP has roughly $T(C+1)$ states. With m resources and capacity vector $(C_1, \dots, C_m)$, the number of capacity states becomes

$$\prod_{i=1}^m (C_i + 1).$$

This is the curse of dimensionality.

6.8 From one resource to a network

Suppose a request consumes several resources at once:

- an itinerary uses seats on multiple flight legs;
- a hotel stay uses a room on several nights;
- an advertisement consumes several campaign budgets;
- a project requires hours from several teams.

The economic logic remains unchanged. For a product consuming resource vector a_j , the exact acceptance condition is

$$r_j \geq v_{t-1}(b) - v_{t-1}(b - a_j).$$

But computing the value function is now prohibitive.

Next lecture we will replace the exact value function by a tractable optimization benchmark. This will produce two different ways to think about decisions:

1. use dual variables as approximate opportunity prices;
2. use a fluid solution as a prediction of what an offline clairvoyant would allocate.

6.9 Takeaways

1. Scarcity makes a current sale interact with future demand.
2. Littlewood's rule compares a sure low fare with the expected marginal value of protected capacity.
3. Dynamic programming makes the opportunity cost exact:
$$\text{accept reward} \geq \text{marginal continuation value.}$$
4. The value of capacity depends on both remaining time and remaining inventory.
5. Exact DP becomes computationally infeasible in resource networks, motivating fluid and offline approximations.

Recommended reading

Core / closest to this lecture.

Talluri and van Ryzin, Section 2.2.1, gives Littlewood's two-class model and the protection-level argument. Sections 2.5.1–2.5.2 give the finite-horizon dynamic formulation and the structure of its optimal policy.

Alternative perspective.

Talluri and van Ryzin, Sections 2.1.1–2.1.2, begins from operational controls—booking limits, protection levels, and displacement cost—before deriving optimization models. Read this for the industry interpretation of the same opportunity-cost logic.

Optional / deeper.

Section 2.2.2 extends the static model to several fare classes. It is useful background for richer capacity-control rules, but the full computational treatment is not required for this lecture.